

RIDESHIELD - ONE-PAGE TECHNICAL PROGRESS SHEET

IoTrix 2.0 Semi-Final | Track A - Embedded IoT System | 5 October 2026

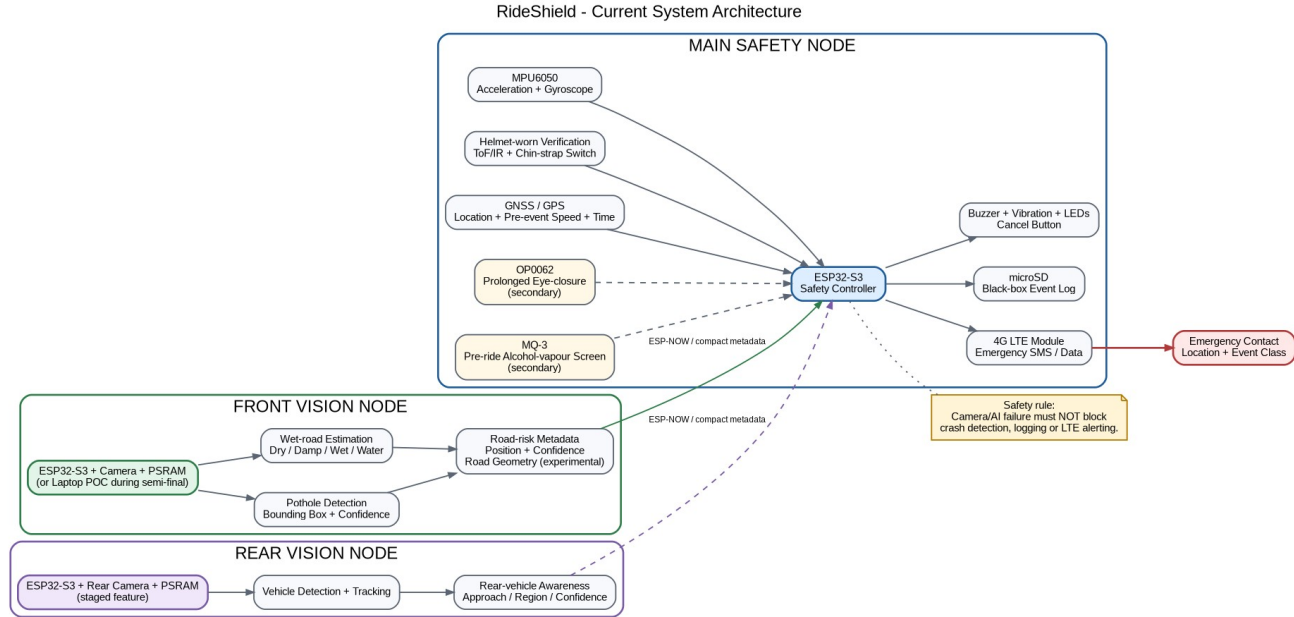
PROBLEM

Motorcycle riders can face delayed emergency response after a crash, road hazards such as potholes and wet surfaces, and limited rear awareness. A safety system must also remain functional when non-critical camera/AI features fail.

PROPOSED SOLUTION

RideShield combines an ESP32-S3 safety node (MPU6050 crash sensing, helmet-worn verification, GPS/LTE, microSD and warnings) with separate front/rear vision nodes. The front vision POC detects potholes and estimates visible road wetness.

ARCHITECTURE



CURRENT PROGRESS

- Current architecture finalized: independent safety core + front/rear vision nodes.
- Pothole CV software workspace/model weights and demo inference pipeline available; quantitative metrics still to be documented.
- Pothole/wet-road dataset and training strategy identified; local Sri Lankan data planned.
- Main hardware interfaces/components selected; crash/LTE/logging integration is the highest-priority build task.
- Rear vision, OP0062 and MQ-3 are staged after core validation.

TECHNOLOGY STACK

- ESP32-S3, MPU6050, ToF/IR + strap switch, GPS, LTE, microSD, cameras.
- C/C++, FreeRTOS-style tasks, UART/I2C/SPI, ESP-NOW.
- Python, OpenCV, lightweight object detection + wet-road classification.

TESTING / VALIDATION

- Semi-final physical test: changed helmet-worn inputs, normal movement vs controlled impact, cancel vs no-cancel.
- Vision test: normal road vs pothole and dry vs visibly wet road samples.
- Record actual attempts, false alarms/misses, SMS latency and resets as soon as measured.
- Do not claim true impact force or measured tyre-road friction.

LIMITATIONS / RISKS

- MPU6050 saturates above ± 16 g; event severity is an index, not a medical diagnosis.
- Helmet camera moves with the rider; geometry/rear zones are confidence-limited.
- LTE current spikes can cause brownouts; dedicated power design is required.
- Vision models require local data for Sri Lankan roads, rain and lighting.

NEXT STEP BEFORE FINAL

- Complete crash -> countdown -> GPS/LTE -> microSD flow and repeat testing.
- Fine-tune and quantify front-vision models; then optimize edge deployment.
- Integrate rear awareness and secondary rider-monitoring features after the core is stable.